Fiber functors for finite symmetric tensor categories in positive characteristic

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Abstract

We prove the finite case of the Benson–Etingof–Ostrik conjecture: every finite symmetric tensor category over an algebraically closed field k of characteristic $p > 0$ admits a k-linear exact faithful strong symmetric monoidal functor to a finite higher Verlinde category $\mathrm{Ver}_{p^n}(k)$. The result covers every prime, including $p = 2$, and the height n may depend on $\mathcal{C}$.

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1 Introduction and main result

What this paper is about, in one paragraph

In characteristic zero, Deligne's theorem says that any "sufficiently nice" symmetric tensor category embeds faithfully into supervector spaces. In positive characteristic there is no single universal target: to catch every finite symmetric tensor category one needs a whole *tower* of finite abelian categories

$$\mathrm{Ver}_p(k) \subset \mathrm{Ver}_{p^2}(k) \subset \mathrm{Ver}_{p^3}(k) \subset \cdots,$$

called the higher Verlinde categories of Benson–Etingof–Ostrik. The claim of Benson–Etingof–Ostrik is that this tower *does* work: every finite symmetric tensor category is a category of representations internal to some finite level of the tower. The present paper proves that claim.

The basic objects

Throughout, k is an algebraically closed field of characteristic $p > 0$.

Definition 1.1 (finite symmetric tensor category)**.** *A* finite symmetric tensor category over k *is a k-linear abelian rigid symmetric monoidal category $\mathcal{C}$ such that:*

(1) the tensor product is biexact;
(2) every $\mathrm{Hom}(X, Y)$ *is finite dimensional over k;*
(3) every object has finite length;
(4) there are finitely many isomorphism classes of simple objects;
(5) the unit object $\mathbf{1}$ *has* $\mathrm{End}(\mathbf{1}) = k$*;*
(6) there are enough projective objects.

Intuition: it is the "linear algebra" world where you can tensor and dualise, everything has finite length, and the unit behaves like the one-dimensional vector space. The simplest example is Vec_k, the category of finite-dimensional k-vector spaces; when $p > 2$ its partner is sVec_k, the $\mathbb{Z}/2$-graded version with the Koszul sign in the symmetry.

Definition 1.2 (fiber functor)**.** *A* fiber functor $\mathcal{C} \to \mathcal{D}$ *between finite symmetric tensor categories is a* k*-linear exact faithful strong symmetric monoidal functor. "Fiber functor" in the classical sense is the special case* $\mathcal{D} = \mathrm{Vec}_k$*; the target being a non-vector-space category is what forces the tower in positive characteristic.*

The higher Verlinde categories

We will use the following facts about $\mathrm{Ver}_{p^n}(k)$, proved in Benson–Etingof–Ostrik (2023).

Lemma 1.3 (properties of the Verlinde tower)**.** *The categories* $\mathrm{Ver}_{p^n}(k)$ *are finite symmetric tensor categories over* k *and they form a tower*

$$\mathrm{Ver}_p(k) \subset \mathrm{Ver}_{p^2}(k) \subset \mathrm{Ver}_{p^3}(k) \subset \cdots,$$

with each inclusion exact and symmetric monoidal. The bottom cases are

$$\mathrm{Ver}_2(k) = \mathrm{Vec}_k, \qquad \mathrm{Ver}_3(k) = \mathrm{sVec}_k, \qquad \mathrm{Ver}_3^+(k) = \mathrm{Vec}_k\,.$$

For every $p^n > 2$ *one has a symmetric tensor subcategory* $\mathrm{Ver}_{p^n}^+(k) \subseteq \mathrm{Ver}_{p^n}(k)$*; concretely* $\mathrm{Ver}_4^+(k)$ *is the category of representations of the connected group scheme* $\alpha_2(k)$ *with its unique symmetric structure, as in Benson–Etingof (2019).*

Intuition for Ver_p: it is the "semisimplification" of $\mathrm{Rep}_k(\mathbb{Z}/p)$: one kills every morphism f with $\mathrm{tr}(gf) = 0$ for all g; the simple objects correspond to Jordan blocks of lengths $1, \ldots, p-1$. For $p = 2$ this collapses to Vec_k; for $p = 3$ it gives supervector spaces. Each higher level Ver_{p^n} is built by abelian-envelope techniques from tilting modules of SL_2.

The main theorem

Here is the result we prove.

Theorem 1.4 (main theorem)**.** *For every prime* p*, every algebraically closed field* k *of characteristic* p*, and every finite symmetric tensor category* $\mathcal{C}$ *over* k*, there exist an integer* $n \geq 1$ *and a fiber functor*

$$\mathcal{C} \longrightarrow \mathrm{Ver}_{p^n}(k).$$

This settles the finite case of Conjecture 1.4 of Benson–Etingof–Ostrik (2023), including $p = 2$. The conjecture in full strength is about categories of *moderate growth*; our theorem is for finite categories, and the finite case already implies the finite part of all the consequences we discuss.

Proof outline

We build a chain of n copies of a functor Φ_n, each time adding a fresh Verlinde factor while forcing the original $\mathcal{C}$ factor to shrink into its small Frobenius-exact part. The four ingredients are:

Step 1. Restricted tilting modulo a Steinberg module. For every finite-dimensional module D over the elementary abelian group $E_r = (\mathbb{Z}/p)^r$, the tensor product $\mathrm{St}_{r-1} \otimes D$ of a fixed Steinberg module with D is a restriction of an SL_2-tilting module. This is the Coulembier–Flake conjecture; we prove it by a vector-bundle argument on the affine plane.

Step 2. Higher Frobenius functor. The previous step lets us define, for every embedding $E_r \hookrightarrow U(k) \subseteq SL_2(k)$, a k-linear additive strong symmetric monoidal functor $\Phi : \mathcal{C} \to \mathcal{C} \boxtimes \mathrm{Ver}_{p^r}(k)$ that is only semilinear ($\Phi(af) = a^{p^r}\Phi(f)$).

Step 3. Exactness by support. For a well-chosen embedding of E_r (with r minimal), Φ is actually exact. The proof uses a projective generator P, the module $M = \mathrm{Hom}(P, P^{\otimes p^r})$, and the theory of the projectivised rank variety.

Step 4. Dimension descent. Even when Φ is only additive (not exact), the Frobenius–Perron dimension constrains composition factors of $\Phi(S)$ for simple S. Iterating Φ finitely many times pushes the first component into the Serre closure of the Frobenius-exact tensor subcategory $\mathcal{C}_{\mathrm{ex}}$.

Four steps then finish: replace the Serre-closure factor by a fixed small Verlinde level (via Etingof–Ostrik for $p > 2$, Etingof–Gelaki for $p = 2$), combine all factors into one big tensor product, kill the scalar twist, and observe that the resulting functor is faithful.

Sections 2–7 carry out the four steps. Section 8 records the standard corollaries.

2 Groundwork

2.1 Simple unit and a projective generator

The definition of finite symmetric tensor category was chosen so that the usual Tannakian toolkit applies. We make this explicit.

Lemma 2.1 ($\mathcal{C}$ is modules over a finite algebra)**.** *Let $\mathcal{C}$ be a finite symmetric tensor category. Then the unit object is simple, and there is a projective generator $P \in \mathcal{C}$. The functor $\omega = \mathrm{Hom}_{\mathcal{C}}(P, -)$ is exact and faithful, and $\mathcal{C}$ is equivalent, as a k-linear abelian category, to the category of finite-dimensional modules over the finite-dimensional k-algebra $A = \mathrm{End}_{\mathcal{C}}(P)^{\mathrm{op}}$.*

Proof. Simple unit is the Deligne–Milne argument. If $U \subset \mathbf{1}$ is a proper subobject with cokernel $V \neq 0$, the epimorphism $U \to V \otimes U$ coming from $\mathbf{1} \twoheadrightarrow V$ has zero composite with the subobject $V \otimes U \twoheadrightarrow V$. So $V \otimes U = 0$. Put $W = \ker(\mathbf{1} \to U^*)$. Then W sits inside a short exact sequence with U, V, and tensoring by W gives $W \simeq V$. The endomorphism ring of the unit has no idempotents, so $V = 0$.

For the projective generator: pick surjections $Q_i \to S_i$ from projectives onto the finitely many simples; set $P = \bigoplus_i Q_i$. Projectivity lifts surjections to P, so every object is a quotient of $P^{\oplus m}$ for some m. Exact faithfulness comes from lifting along $P \to$ image. To identify $\mathcal{C} \simeq A$-mod, note that presenting objects by copies of P and using projectivity is fully faithful on the level of presentations, and every finite A-module has a finite free presentation. □

We fix such a P and such an A once and for all. In particular, $P \twoheadrightarrow \mathbf{1}$, since the unit is one of the simples.

2.2 Deligne products

For finite k-linear abelian categories $\mathcal{A}, \mathcal{B}$, the *Deligne product* $\mathcal{A} \boxtimes \mathcal{B}$ is the universal recipient of functors that are k-bilinear and right exact in each variable. When $\mathcal{A} \simeq A$-mod and $\mathcal{B} \simeq B$-mod, we have $\mathcal{A} \boxtimes \mathcal{B} \simeq (A \otimes_k B)$-mod, and its simple objects are the exterior products of the simples of the two factors (here we use k algebraically closed).

Lemma 2.2 (combining exact tensor functors)**.** *Let $\mathcal{A}_1, \dots, \mathcal{A}_s, \mathcal{Z}$ be finite symmetric tensor categories and let $F_i : \mathcal{A}_i \to \mathcal{Z}$ be exact k-linear strong symmetric monoidal functors. Then*

$$H : \mathcal{A}_1 \boxtimes \cdots \boxtimes \mathcal{A}_s \to \mathcal{Z}, \qquad H(X_1 \boxtimes \cdots \boxtimes X_s) = F_1(X_1) \otimes \cdots \otimes F_s(X_s),$$

extends uniquely to an exact k-linear strong symmetric monoidal functor.

Proof. The displayed multilinear functor has a right-exact extension by the universal property of the Deligne product. Symmetry in $\mathcal{Z}$ gives the tensor constraint by rearranging factors, and coherence extends by the same universal property. The result preserves duals. A short exact sequence dualises, is sent to a right-exact sequence, and is dualised back; hence the functor is also left exact. □

2.3 Frobenius–Perron dimension

For finite tensor categories the Grothendieck ring $K_0(\mathcal{C})$ is finitely freely generated by simple classes. Symmetry makes it a commutative ring, and the Frobenius–Perron theorem identifies a unique positive normalised real character $d = \mathrm{FPdim} : K_0(\mathcal{C}) \to \mathbb{R}$ with $d(\mathbf{1}) = 1$ and d multiplicative on classes. Key facts, all in Etingof– Gelaki–Nikshych–Ostrik (2015):

Lemma 2.3 (FP dimension toolkit)**.** *Let $\mathcal{C}$ be a finite tensor category.*

(i) $d(X) \geq 1$ *for every nonzero X; equality if and only if X is invertible.*
(ii) If $0 \to X \to Y \to Z \to 0$ *is exact, then* $d(Y) = d(X) + d(Z)$.
(iii) On a Deligne product, $d(A \boxtimes B) = d(A)\, d(B)$.
(iv) If $F : \mathcal{C} \to \mathcal{D}$ *is an exact strong monoidal functor of finite tensor categories, then* $d(F(X)) = d(X)$ *for every X.*

Beware: the categorical dimension $\dim_{\mathrm{cat}}(B) \in k$ that appears in section 6 is a different, k-valued invariant; it is *not* Frobenius– Perron dimension. We will only use categorical dimension for one mod-p congruence.

3 Step 1: Restricted tilting modules

3.1 The statement

Let $G = SL_2(k)$, and let $U \simeq \mathbb{G}_a$ be its upper unipotent subgroup, identified with k via $u(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$. For $r \geq 1$ put $E_r = (\mathbb{Z}/p)^r$ and let $\iota : E_r \hookrightarrow U(k)$ be any injective homomorphism. Write $R = R_\iota$ for the restriction of G-modules along ι.

Let $L(h)$ denote the simple rational G-module of highest weight $h \geq 0$; let $\mathrm{St}_m = L(p^m - 1)$ be the m-th *Steinberg module*, so $\mathrm{St}_0 = k$. Let $\nabla(a) = \mathrm{Sym}^a(V^*)$ be the a-th costandard module, where $V = k^2$ is the standard representation, and $\Delta(a) = \nabla(a)^*$. A G-module is *tilting* if it has both a costandard filtration and a Weyl filtration; $T(a)$ is the indecomposable tilting module of highest weight a.

Theorem 3.1 (restricted-tilting theorem)**.** *Let $r \geq 1$, $E_r = (\mathbb{Z}/p)^r$, and $\iota : E_r \to U(k)$ any injective homomorphism. For every finite-dimensional kE_r-module D there is a tilting G-module T such that*

$$R_\iota(\mathrm{St}_{r-1}) \otimes D \cong R_\iota(T)$$

as kE_r-modules.

This proves Conjecture 5.4.2(1) of Coulembier–Flake (2026) and Conjecture 1 of Coulembier–Flake (2025), for every prime.

3.2 The plan of the proof

We do three things:

(a) lift the arbitrary kE_r-module D to a coherent rational U-module $\widehat{D}$ of bounded "degree" (a bound on polynomial degree of the unipotent action matrix);

(b) use $\widehat{D}$ to construct a G-equivariant vector bundle $\mathcal{M}$ on the plane $\mathbb{A}^2_k$ whose origin fibre has all simple composition factors below a height bound;

(c) tensor by St_{r-1}, observe the resulting bundle's origin fibre is tilting, and compare origin fibre with e_1-fibre via a formal construction on jets.

We then recover D from the e_1-fibre.

3.3 Lifting D to a bounded U-module

For a finite-dimensional rational U-module D, its *degree* is the largest exponent of t appearing in the action matrix $\rho_D(u(t))$; this does not depend on a chosen basis.

Proposition 3.2 (bounded lift)**.** *For every finite-dimensional kE_r-module D, there is a finite-dimensional rational U-module $\widehat{D}$ of degree $< p^r$ such that $\widehat{D}|_{E_r} \cong D$ along ι.*

Proof. Let $H \subseteq k$ be the image of E_r under ι. The vector space $C_q = \mathrm{span}_k\{1, t, \dots, t^{q-1}\} \subseteq k[t]$ (with $q = p^r$) is a subcoalgebra of $k[t]$ with $\Delta f(t,s) = f(t+s)$. Evaluation at H gives a coalgebra map $\mathrm{ev}_H : C_q \to k^H = \mathcal{O}(E_r)$, which is a bijection: the two sides have dimension q, and a degree-$< q$ polynomial vanishing at the q distinct points of H is zero. Transporting the $\mathcal{O}(E_r)$-coaction defining D across ev_H, then including $C_q \subset k[t]$, gives a rational U-module of degree $< q$. Its restriction is D. □

3.4 The G-equivariant bundle on the plane

Now consider the orbit map $\pi : G \to \mathbb{A}^2 \setminus \{0\}$, $g \mapsto ge_1$, with $e_1 = (1,0)$. This is a right U-torsor: the fibres are right U-cosets and on the standard opens $x \neq 0$, $y \neq 0$ it has obvious rational sections. Form the associated vector bundle

$$\mathcal{E} = G \times^U \widehat{D}, \qquad (g,d) \sim (gu, u^{-1}d),$$

on the punctured plane, with G acting by left multiplication. The fibre at e_1 is $\widehat{D}$ itself.

Proposition 3.3 (extension and weight bound)**.** *The bundle $\mathcal{E}$ extends to a G-equivariant vector bundle $\mathcal{M}$ on the whole affine plane, and every simple composition factor of the origin fibre $M_0 = \mathcal{M}|_0$ has highest weight $< q$.*

Proof. Extension. A coherent extension $\mathcal{F}$ exists on the noetherian affine plane. Its double dual $\mathcal{M} = \mathcal{F}^{**}$ is a coherent reflexive sheaf restricting to $\mathcal{E}$. On the regular surface $\mathbb{A}^2$ every reflexive coherent sheaf is locally free: at a closed point the stalk has depth two, so the Auslander–Buchsbaum formula gives projective dimension zero, and at the generic point and at the generic point of a curve the stalks are trivially free. Hence $\mathcal{M}$ is a vector bundle.

Equivariance. Over the punctured plane we have a G-equivariant bundle isomorphism $\pi_2^*\mathcal{M} \cong a^*\mathcal{M}$ for the action $a : G \times \mathbb{A}^2 \to \mathbb{A}^2$ and projection $\pi_2 : G \times \mathbb{A}^2 \to \mathbb{A}^2$. Both sides are locally free, so the Hom-sheaf is locally free and its sections extend across the codimension-two subset $G \times \{0\}$; the extensions of the isomorphism and its inverse remain inverse because the two composites agree on the punctured plane and hence, by uniqueness, everywhere. The identity and cocycle conditions follow the same way.

Weight bound at the origin. The line $\ell = ke_1$ is fixed pointwise by U. For $a \in k^\times$ put $h_a = \mathrm{diag}(a, a^{-1})$; the bundle action of h_a gives a linear identification of fibres at e_1 and at ae_1, and $h_a^{-1}u(t)h_a = u(a^{-2}t)$. Hence on every nonzero fibre over ℓ the U-action has degree $< q$. Let $A = \Gamma(\ell, \mathcal{E}nd(\mathcal{M}|_\ell))$ and write the U-action as an element of $A \otimes_k k[t]$, $\sum_j b_j t^j$. For $j \geq q$, b_j vanishes on every nonzero fibre, so $b_j = 0$ (since k is infinite and A is torsion-free over $k[t]$). So the origin action also has degree $< q$, and the same holds for any subquotient.

Passing degree to weights. Write a simple G-module as a tensor product of Frobenius twists of $\mathrm{Sym}^{h_i}(k^2)$ with $h = \sum_i h_i p^i$, $0 \leq h_i < p$. Its U-action has degree exactly h (the coefficient of t^h from $\bigotimes_i e_2^{h_i}$ to $\bigotimes_i e_1^{h_i}$ is nonzero). Hence every simple composition factor $L(h)$ of M_0 has $h < q$. □

3.5 Tilting under a height bound

We use two standard facts about SL_2-tilting modules, both due to Donkin and both valid in every positive characteristic including $p = 2$:

$$L(a) = T(a) \quad (0 \leq a < p), \qquad T(a) \otimes T(b)^{(1)} \simeq T(a + pb) \quad (p - 1 \leq a \leq 2p - 2,\ b \geq 0), \tag{3.1}$$

where (1) means pullback along the Frobenius. See Benson– Etingof–Ostrik (2023), Proposition 3.1.

Lemma 3.4 (Steinberg ⊗ bounded weight is tilting)**.** *Let M be a finite-dimensional rational $SL_2(k)$-module with every composition factor of the form $L(h)$ for $0 \leq h < p^m$ (where $m \geq 1$). Then $\mathrm{St}_{m-1} \otimes M$ is tilting.*

Proof. Reduces by composition series to $M = L(h)$.

For $m = 1$, $h < p$ and $\mathrm{St}_0 = k$, so $\mathrm{St}_0 \otimes L(h) = T(h)$ is tilting by (3.1).

For $m > 1$, write $h = d + ph'$ with $0 \leq d < p$ and $h' < p^{m-1}$. Steinberg's tensor product theorem gives

$$\mathrm{St}_{m-1} \otimes L(h) \cong L(d) \otimes \mathrm{St}_1 \otimes \big(\mathrm{St}_{m-2} \otimes L(h')\big)^{(1)}.$$

By induction the bracket is a direct sum of tilting modules. Apply Donkin's formula (3.1) with $a = p-1$ and b equal to the highest weight of each summand; each summand times St_1 (a Frobenius twist of a tilting) is tilting. Tensoring further with $L(d) = T(d)$ preserves tilting because tilting modules are closed under tensor products. □

Applying lemma 3.4 with $m = r$ to $M_0 = \mathcal{M}|_0$, we have:

$$T := \mathrm{St}_{r-1} \otimes M_0 \tag{3.2}$$

is a tilting G-module. The remaining step is to compare T (the origin fibre of $\mathcal{N} = \mathrm{St}_{r-1} \otimes_k \mathcal{M}$) with the e_1-fibre of $\mathcal{N}$.

3.6 Formal comparison of fibres

Lemma 3.5 (comparison with a tilting origin fibre)**.** *Let $\mathcal{N}$ be a G-equivariant vector bundle on $\mathbb{A}^2$ whose origin fibre $T = \mathcal{N}|_0$ is tilting. Then $\mathcal{N}|_{e_1} \cong T|_U$ as rational U-modules.*

Proof. Jet trivialisation. Let $A = k[x, y]$, $\mathfrak{m} = (x, y)$, and $N = \Gamma(\mathbb{A}^2, \mathcal{N})$, a possibly infinite-dimensional rational G-module. Local freeness gives canonical G-isomorphisms

$$\mathfrak{m}^j N / \mathfrak{m}^{j+1} N \cong \mathrm{Sym}^j(V^*) \otimes_k T. \tag{3.3}$$

The target has a good filtration. Tensor products of Sym's of V^* have good filtrations: via multiplication by the determinant $d = x_1y_2 - y_1x_2$, the sequence $0 \to \mathrm{Sym}^{a-1} V^* \otimes \mathrm{Sym}^{b-1} V^* \to \mathrm{Sym}^a V^* \otimes \mathrm{Sym}^b V^* \to \mathrm{Sym}^{a+b} V^* \to 0$ is exact.

Now T has both a Weyl and a good filtration, so for every j,

$$\mathrm{Ext}^1_G(T, \mathrm{Sym}^j(V^*) \otimes T) = 0$$

by Cline–Parshall–Scott–van der Kallen.

Starting from the identity $s_1 : T \to N/\mathfrak{m}N$, we lift successively to G-maps $s_j : T \to N/\mathfrak{m}^j N$ that reduce modulo $\mathfrak{m}^j$ to s_{j-1}; the obstruction is exactly the Ext group just computed. Extending scalars, s_j gives an $A/\mathfrak{m}^j$-linear map $(A/\mathfrak{m}^j) \otimes_k T \to N/\mathfrak{m}^j N$, which is an isomorphism by Nakayama (both sides free of the same rank and the map is an isomorphism on the residue field). The inverse limit is a G-equivariant isomorphism $k[[x, y]] \otimes_k T \xrightarrow{\sim} \widehat{N}$.

Fibre comparison. Restrict to the fixed line $y = 0$, pass to $K = k((x))$: this gives a U_K-isomorphism $\varphi : T_K \cong \mathcal{N}_{xe_1}$. To cancel the parameter mismatch we use the torus element $h = \mathrm{diag}(x, x^{-1})$: since $hu(t)h^{-1} = u(x^2t)$, the composite $f = b_h^{-1}\varphi\rho_T(h)$ intertwines the original U-actions (where $b_h : \mathcal{N}_{e_1} \to \mathcal{N}_{xe_1}$ is bundle transport).

Descent to k. The space $L = \mathrm{Hom}_U(T, \mathcal{N}_{e_1})$ is cut out by finitely many k-linear equations in fixed k-bases, so $L_K = L \otimes_k K$. The determinant on L is nonzero over K (witnessed by f), so nonvanishing at some k-point (since k is infinite) gives the required isomorphism over k. □

Proof of theorem 3.1. Apply proposition 3.3 to get $\mathcal{M}$ with origin fibre M_0 whose simple factors have highest weight $< q = p^r$. By lemma 3.4, $T = \mathrm{St}_{r-1} \otimes M_0$ is tilting. Apply lemma 3.5 to $\mathcal{N} = \mathrm{St}_{r-1} \otimes_k \mathcal{M}$: its origin fibre is T, so

$$\mathcal{N}|_{e_1} \cong T|_U.$$

But by construction $\mathcal{N}|_{e_1}$ restricts to $R(\mathrm{St}_{r-1}) \otimes D$ along $E_r \hookrightarrow U(k)$, hence so does T. Set $T_{\mathrm{out}} := T$; the theorem follows. □

4 Step 2: The higher Frobenius functor

4.1 The functor Θ from group modules to Verlinde

Fix $r \geq 1$, $q = p^r$, and an embedding $E = E_r \hookrightarrow U(k)$. We write R for restriction of SL_2-modules along this embedding, and $\mathcal{W} = \mathrm{Ver}_{p^r}(k)$. Let $\mathcal{T} \subseteq \mathrm{Rep}_k(SL_2)$ be the category of tilting modules, and let I_r be the tensor ideal generated by $\mathrm{St}_r = L(p^r - 1)$.

Concretely, I_r is defined by demanding that $T(a)$ (the indecomposable tilting module of highest weight a) vanish in $\mathcal{T}/I_r$ whenever $a \geq p^r - 1$; this is the standing convention in this circle of papers.

The functor $\mathcal{T}/I_r \to \mathcal{W}$ is an equivalence onto a full subcategory (Benson–Etingof–Ostrik (2023), Theorem 4.2(i)). Stable restriction gives a faithful quotient functor $\mathcal{T}/I_r \to \mathrm{Stab}(E) := \mathrm{Rep}_k(E)/(\text{projectives})$.

Coulembier–Flake (2026, Theorem 5.4.3) state: *if for every finite-dimensional E-module D the tensor $R(\mathrm{St}_{r-1}) \otimes D$ is a restriction of a tilting, then there is a k-linear strong symmetric monoidal functor*

$$\Theta : \mathrm{Rep}_k(E) \longrightarrow \mathcal{W} \quad \text{with} \quad \Theta(R(T)) \simeq \Sigma_r(T \bmod I_r). \tag{4.1}$$

The hypothesis is exactly theorem 3.1. So theorem 3.1 produces Θ, once and for all embeddings.

Θ is not exact; it is additive and symmetric monoidal.

Lemma 4.1 (Θ detects nonprojectives)**.** *Let T be a tilting module whose restriction $R(T)$ is not projective as an E-module. Then $\Theta(R(T)) \neq 0$. Moreover, if $H < E$ is a proper subgroup, then every module induced from a kH-module is killed by Θ.*

Proof. If $T \bmod I_r = 0$ then $R(T)$ is projective (stable restriction result). Otherwise $T \bmod I_r \neq 0$, and Σ_r being fully faithful implies $\Theta(R(T)) \neq 0$.

For induced modules: $R(\mathrm{St}_r)$ has dimension q, and its restriction is projective (the mod I_r quotient kills St_r). Since kE is local, a projective of dimension q is the free module, so Θ kills every projective. When $H < E$ is proper, $\mathrm{St}_{r-1} \in I_s$ at the level $s = \mathrm{rank}(H)$ so $\mathrm{Res}^E_H R(\mathrm{St}_{r-1})$ is projective; the tensor identity $S \otimes \mathrm{Ind}^E_H D \cong \mathrm{Ind}^E_H(\mathrm{Res}^E_H S \otimes D)$ therefore gives a projective E-module, on which Θ vanishes. Since $\Theta(S) \neq 0$ and tensoring by a nonzero object is faithful, $\Theta(\mathrm{Ind}^E_H D) = 0$. □

4.2 Internal evaluation

Let $\mathcal{C}$ be a finite symmetric tensor category with projective generator P_0. Write $A = \mathrm{End}_{\mathcal{C}}(P_0)^{op}$, so $\mathcal{C} \simeq A\text{-mod}$. An object of $\mathcal{C} \boxtimes \mathrm{Rep}_k(E)$ is an object of $\mathcal{C}$ equipped with a commuting E-action. Applying Θ to the underlying E-module and to the action maps produces an A-module object in $\mathcal{W}$, hence an object of $\mathcal{C} \boxtimes \mathcal{W}$. This is the *internal extension* $\Theta_{\mathcal{C}}$ of Θ (Coulembier–Flake 2026, Example 3.2.4 and Proposition 3.2.6; requires only additivity of Θ, not exactness).

Lemma 4.2 (evaluation at projectives)**.** *For $Q \in \mathcal{C}$ projective and Z an E-object of $\mathcal{C}$,*

$$(\mathrm{Hom}_{\mathcal{C}}(Q, -) \boxtimes \mathrm{Id}_{\mathcal{W}})(\Theta_{\mathcal{C}}(Z)) \cong \Theta(\mathrm{Hom}_{\mathcal{C}}(Q, Z)).$$

Proof. In A-mod, Q is a direct summand of a free module, so $\mathrm{Hom}_A(Q, -)$ is an additive idempotent of the identity functor on A-mod. An additive functor preserves split idempotents and direct sums; applying Θ therefore commutes with $\mathrm{Hom}_A(Q, -)$. □

4.3 The tensor-power functor Φ

Give $X^{\otimes q}$ the E-action coming from regular permutation of tensor positions. Define

$$\Phi(X) = \Theta_{\mathcal{C}}(X^{\otimes q}) \in \mathcal{C} \boxtimes \mathcal{W}. \tag{4.2}$$

Lemma 4.3 (monoidal structure and semilinearity)**.** *Φ is additive, strong symmetric monoidal, strong k-linear after twisting scalars by Frobenius $a \mapsto a^q$, and satisfies $\Phi(af) = a^q \Phi(f)$ for scalars $a \in k$.*

Proof. The monoidal constraint comes from the E-equivariant reordering $(X \otimes Y)^{\otimes q} \cong X^{\otimes q} \otimes Y^{\otimes q}$ with the diagonal action on the right, applied through the internal monoidal extension of Coulembier–Flake (2026, Section 3.3.2). Additivity can be checked directly: in the expansion of $(X \oplus Y)^{\otimes q}$, a nonconstant colouring of tensor positions has a proper stabiliser in E, so the sum over its E-orbit is induced from that stabiliser and hence is killed by Θ (using lemma 4.1); only the two constant colourings survive. Semilinearity is immediate from the same calculation on scalars. □

Combining equation (4.2) and lemma 4.2:

$$(\mathrm{Hom}_{\mathcal{C}}(Q, -) \boxtimes \mathrm{Id}_{\mathcal{W}})(\Phi(X)) \cong \Theta(\mathrm{Hom}_{\mathcal{C}}(Q, X^{\otimes q})) \tag{4.3}$$

for every projective Q.

4.4 The nonprojective image criterion

Proposition 4.4 (Φ exact iff image nonzero)**.** *If there are projective objects $P, Q \in \mathcal{C}$ with $\Theta(\mathrm{Hom}_{\mathcal{C}}(Q, P^{\otimes q})) \neq 0$, then Φ is exact.*

Proof. Equation (4.3) gives $\Phi(P) \neq 0$. For a short exact sequence in $\mathcal{C}$, tensoring by P splits it because tensor products of projectives with arbitrary objects are projective. The additive Φ preserves this split sequence. Monoidality identifies the result as the original image sequence tensored by $\Phi(P)$. Tensoring by the nonzero $\Phi(P)$ is exact and faithful, and reflects exactness; the image sequence is exact. □

5 Step 3: Choosing an exact embedding

Now we need to actually find r and ι for which the criterion of proposition 4.4 fires. The strategy: take r minimal so that $M(r) := \mathrm{Hom}_{\mathcal{C}}(P, P^{\otimes p^r})$ (as a kE_r-module via the regular permutation of tensor slots) is nonprojective; and then choose the embedding ι so that $R(\mathrm{St}_{r-1}) \otimes M(r)$ stays nonprojective.

5.1 Norms and p-group projectivity

For a finite group H acting on a finite-dimensional vector space, $\nu_H : X_H \to X^H$ is the *norm map*: it factors $\sum_{h \in H} h : X \to X$.

Lemma 5.1 (p-group projectivity via norm)**.** *Let k be a field of characteristic p, H a finite p-group, M a finite-dimensional kH-module. Then M is projective iff $\nu_H(M)$ is an isomorphism. More generally, for any finite group G, the norm map on a projective kG-module is an isomorphism.*

Proof. The group ring kH is local self-injective: its only simple is k with trivial action; its socle is generated by $N_H = \sum_{h \in H} h$; the augmentation ideal is nilpotent (induct on $|H|$, use a central order-p element z with $(z-1)^p = 0$, descend to $k[H/\langle z \rangle]$); and a nondegenerate pairing $ab \mapsto$ coefficient of 1 identifies kH with its linear dual as a module.

For kH-free modules, the norm map is an iso. Converse: if $N_H m \neq 0$, the map $kH \to M$, $a \mapsto am$ is injective and splits by self-injectivity, so M has a free summand. Strip off all free summands, leaving M' with $N_H M' = 0$; if $M' \neq 0$ then $N_H = M'_H = 0$ by Nakayama, contradiction with an iso. So $M' = 0$.

For a general finite group and a projective module: restrict to Sylow p-subgroups; each restriction is projective; the norm iso there implies the total norm iso (same argument). □

Lemma 5.2 (independent actions multiply)**.** *Let finite p-groups H_i act on objects $X_i \in \mathcal{C}$ ($i = 1, \ldots, s$). If $\omega(X_i)$ is projective over kH_i for exact faithful $\omega = \mathrm{Hom}_{\mathcal{C}}(P, -)$, then $\omega(X_1 \otimes \cdots \otimes X_s)$ is projective over $k[H_1 \times \cdots \times H_s]$ (independent actions), and over every subgroup thereof.*

Proof. Categorically, $(X \otimes Y)_{H \times K} \cong X_H \otimes Y_K$ and $(X \otimes Y)^{H \times K} \cong X^H \otimes Y^K$ (take kernels/cokernels one group at a time). So $\nu_{H \times K}(X \otimes Y) = \nu_H(X) \otimes \nu_K(Y)$. If each $\nu_{H_i}(X_i)$ is iso, so is the product norm. Applying exact ω and lemma 5.1 finishes. Restrict to a subgroup $H \leq H_1 \times \cdots \times H_s$: kH_i is free over kH for any finite group, so projectivity is preserved. □

5.2 Existence of a minimal r

Let $E_a = (\mathbb{Z}/p)^a$ act on p^a positions by its regular representation; this induces an E_a-action on $P^{\otimes p^a}$. Set $M(a) = \omega(P^{\otimes p^a})$, an E_a-module.

Proposition 5.3 (first nonprojective power)**.** *There is a least integer $r \geq 1$ for which $M(r)$ is not projective over kE_r. For this r, the restriction of $M(r)$ to every proper subgroup of E_r is projective.*

Proof. Suppose all $M(a)$ projective. Fix d and an elementary abelian p-subgroup $H \leq \mathfrak{S}_d$. Every H-orbit in $\{1, \dots, d\}$ is regular because H is abelian; orbitwise, $P^{\otimes d}$ is a product of $P^{\otimes p^{a_i}}$, each projective over kE_{a_i} by assumption. By lemma 5.2, $\omega(P^{\otimes d})$ is projective over kH for the product of orbit groups, hence over the diagonal copy of H (which is an injective image). Chouinard's theorem then says projective over every elementary abelian p-subgroup implies projective over $k\mathfrak{S}_d$.

Now apply $P \twoheadrightarrow \mathbf{1}$ ($\mathfrak{S}_d$-equivariant, trivial action on the target) and exactness of ω followed by coinvariants:

$$\omega(P^{\otimes d})_{\mathfrak{S}_d} \twoheadrightarrow \omega(\mathbf{1}) \neq 0.$$

By lemma 5.1, the total symmetrizer $b_d = \sum_{\sigma \in \mathfrak{S}_d} \sigma$ acts nontrivially on $P^{\otimes d}$.

On the other hand the action map $k\mathfrak{S}_d \to \mathrm{End}_{\mathcal{C}}(P^{\otimes d})$ cannot be injective: length grows exponentially in d (bounded by $\ell(P)b^{d-1}$ with $b = \max_i \ell(S_i \otimes P)$) and the dimension of $\mathrm{Hom}(X, Y)$ is bounded by $\ell(X)\ell(Y)$, whereas $d!$ grows like a factorial. The kernels of these action maps constitute a tensor ideal in the permutation category, and every nonzero tensor ideal in Sym contains some b_d in characteristic $p > 0$ (Coulembier–Etingof–Ostrik 2023, Lemma 4.10). Its proof uses Kleshchev's faithfulness theorem. Contradiction. So some $M(r)$ is nonprojective; take r minimal.

For the last claim: $H < E_r$ of rank $a < r$ acts on the regular E_r-set as a disjoint union of regular H-orbits; each orbit's power-evaluation is projective by minimality (rank a) or trivially if $H = 1$; apply lemma 5.2 and restrict to the diagonal. □

5.3 Rank varieties and support

Let kE_r-modules be graded by the augmentation ideal $\mathfrak{m} = (g_i - 1)_i$. The *projectivised rank variety* of a finite-dimensional module D is

$$\mathrm{supp}(D) = \{[\alpha] \in \mathbb{P}^{r-1}(k) : D \text{ not free over } k[t]/(t^p),\ t \mapsto \textstyle\sum_i \alpha_i(g_i - 1)\}.$$

Two standard facts (Benson 2017; Coulembier–Flake 2025):

$$\mathrm{supp}(D) = \varnothing \iff D \text{ projective}, \qquad \mathrm{supp}(D \otimes D') = \mathrm{supp}(D) \cap \mathrm{supp}(D'). \tag{5.1}$$

Lemma 5.4 (support point independence)**.** *If M is a nonprojective kE_r-module with all restrictions to proper subgroups projective, then every $[\alpha] \in \mathrm{supp}(M)$ has $\mathbb{F}_p$-linearly independent coordinates.*

Proof. For $c \in (\mathbb{F}_p^r)^*$, let $H = \ker c$ and $N = \mathrm{Ind}_H^{E_r} k$, the regular representation of $E_r/H \cong \mathbb{Z}/p$ (identified with $k[z]/(z^p)$, generator acts by $1 + z$). The element $\sum_i \alpha_i(g_i - 1)$ acts on N by $h_\alpha(z) = \sum_i \alpha_i((1+z)^{c_i} - 1)$, whose linear coefficient is $\sum_i c_i\alpha_i$.

If $\sum_i c_i\alpha_i \neq 0$, then $t \mapsto h_\alpha(z)$ is an iso of truncated polynomial algebras, so N is free as $k[t]/(t^p)$; if $\sum_i c_i\alpha_i = 0$, then $h_\alpha \in (z^2)$, so $h_\alpha^{p-1} = 0$, incompatible with a free module of dimension p; so N is not free.

Thus $\operatorname{supp}(N) = \{[\alpha] : \sum_i c_i\alpha_i = 0\}$. The tensor identity gives $M \otimes \operatorname{Ind}_H^{E_r} k \cong \operatorname{Ind}_H^{E_r}(M|_H)$, which is projective, so $\operatorname{supp}(M) \cap \operatorname{supp}(N) = \varnothing$. Hence $\sum_i c_i\alpha_i \neq 0$ for every nonzero c. In particular the coordinates of every $[\alpha] \in \operatorname{supp}(M)$ are $\mathbb{F}_p$-linearly independent. □

Lemma 5.5 (Steinberg support in coordinates)**.** *Let $\lambda_1, \dots, \lambda_r \in k$ be $\mathbb{F}_p$-linearly independent and define $\iota_\lambda : E_r \to SL_2(k)$, $g_i \mapsto u(\lambda_i)$. Write R_λ for restriction. Then*

$$\operatorname{supp}(R_\lambda(\mathrm{St}_{r-1})) = \left\{ [\alpha] : \sum_i \alpha_i \lambda_i^{p^j} = 0 \quad (0 \le j \le r-2) \right\}. \tag{5.2}$$

Proof. On $\mathrm{St}_1 = \mathrm{Sym}^{p-1}(k^2)$, the unipotent $u(a)$ acts as $\exp(aN) = \sum_{b=0}^{p-1} a^b N^b/b!$ where N is a $p \times p$ Jordan block. Then $\sum_i \alpha_i(g_i - 1)$ acts by a polynomial in N whose linear coefficient is $\sum_i \alpha_i\lambda_i$; by the argument of lemma 5.4, the resulting p-dimensional module is free iff the coefficient is nonzero.

Frobenius twist replaces λ_i by $\lambda_i^{p^j}$. Since $\mathrm{St}_{r-1} \cong \bigotimes_{j=0}^{r-2} \mathrm{St}_1^{(j)}$, the tensor product formula (5.1) gives an intersection of hyperplanes $M_j : \sum_i \alpha_i\lambda_i^{p^j} = 0$, exactly equation (5.2). For $r = 1$ the system is empty and $\mathrm{St}_0 = k$ has $\operatorname{supp}(\mathrm{St}_0) = \mathbb{P}^0(k)$, consistent. □

5.4 Moore's lemma: choosing λ

Lemma 5.6 (λ solving prescribed equations)**.** *Let $\alpha_1, \dots, \alpha_r \in k$ be $\mathbb{F}_p$-linearly independent. Then there exist $\mathbb{F}_p$-linearly independent $\lambda_1, \dots, \lambda_r \in k$ with*

$$\sum_i \alpha_i \lambda_i^{p^j} = 0, \qquad 0 \le j \le r-2. \tag{5.3}$$

For $r = 1$ the system is empty.

Proof. $r = 1$: take $\lambda_1 \neq 0$. For $r \ge 2$: step is standard graded linear algebra. Put $\beta_i = \alpha_i^{p^{-(r-2)}}$ (using that k is perfect). Let $B = (\beta_i^{p^s})_{s,i}$ be the $(r-1) \times r$ matrix.

Rows of B are linearly independent: otherwise a nontrivial additive polynomial $f(X) = \sum_{s=0}^{r-2} a_s X^{p^s}$ vanishes on the p^r-element span of the β_i, but $\deg f \le p^{r-2}$, so $f = 0$.

Row space of B contains no nonzero $\mathbb{F}_p^r$-vector: if $c_i = f(\beta_i) \in \mathbb{F}_p$, then the polynomial $f^p - f$ vanishes on the same p^r-element set and has degree $\le p^{r-1}$; so $f^p - f = 0$, forcing $f = 0$ (since $\deg f < p^{r-1}$ and f has zero constant term).

Choose $\lambda \in \ker B$ nonzero. Kernel is one-dimensional. Any $c \in \mathbb{F}_p^r$ with $\sum_i c_i\lambda_i = 0$ belongs to the annihilator of $\ker B$, i.e., to the row space of B, hence is $c = 0$. So λ_i are $\mathbb{F}_p$-linearly independent. Reading rows of B in reverse order, $(\alpha_i^{p^{-j}})_i$, and raising the equations $\sum_i \lambda_i^{p^s} \alpha_i^{p^{-(r-2)}p^s} = 0$ to the p^{r-2}-th power gives (5.3). □

Proposition 5.7 (embedding kills no support)**.** *Let M be a nonprojective kE_r-module with projective restriction to every proper subgroup. Then there is an embedding $\iota_\lambda : E_r \hookrightarrow U(k)$ with $R_\lambda(\mathrm{St}_{r-1}) \otimes M$ nonprojective.*

Proof. Pick $[\alpha] \in \operatorname{supp}(M)$; coordinates are $\mathbb{F}_p$-linearly independent by lemma 5.4. Choose λ by lemma 5.6; lemma 5.5 then gives $[\alpha] \in \operatorname{supp}(R_\lambda(\mathrm{St}_{r-1}))$. So the tensor product support (5.1) is nonempty. Hence $R_\lambda(\mathrm{St}_{r-1}) \otimes M$ is nonprojective. □

5.5 Exactness

Theorem 5.8 (exact higher Frobenius functor)**.** *Let $\mathcal{C}$ be a finite symmetric tensor category over k of characteristic $p > 0$. Then there exist $r \geq 1$ and an embedding $E_r \hookrightarrow U(k)$ such that the construction of section 4 gives an exact additive strong symmetric monoidal functor*

$$\Phi : \mathcal{C} \to \mathcal{C} \boxtimes \mathrm{Ver}_{p^r}(k)$$

with $\Phi(X) = \Theta_{\mathcal{C}}(X^{\otimes p^r})$ and $\Phi(af) = a^{p^r}\Phi(f)$. Moreover, for every projective Q the evaluation (4.3) holds.

Proof. Let P be the projective generator and r the minimal integer of proposition 5.3. Then $M = \mathrm{Hom}_{\mathcal{C}}(P, P^{\otimes p^r})$ is nonprojective with projective restriction to every proper subgroup of E_r. By proposition 5.7 pick λ so that $R_\lambda(\mathrm{St}_{r-1}) \otimes M$ is nonprojective. By theorem 3.1, $R_\lambda(\mathrm{St}_{r-1}) \otimes M \cong R_\lambda(T)$ for a tilting module T. By lemma 4.1,

$$0 \neq \Theta(R_\lambda(T)) \cong \Theta(R_\lambda(\mathrm{St}_{r-1})) \otimes \Theta(M).$$

So $\Theta(M) \neq 0$. Proposition 4.4 with $P = Q$ shows Φ is exact. □

6 Step 4: Dimension descent

We now have an exact $\Phi : \mathcal{C} \to \mathcal{C} \boxtimes \mathcal{W}$, $q = p^r$, $\mathcal{W} = \mathrm{Ver}_{p^r}(k)$, semilinear. Its image closes up on the first factor to the Serre closure of a Frobenius-exact subcategory.

6.1 Ordinary Frobenius and a mod-p identity

For $Y \in \mathcal{C}$, let c be the cyclic permutation of the p factors of $Y^{\otimes p}$ and $D = 1 - c$ (so $D^p = 0$). Define the Etingof–Ostrik components:

$$F_i(Y) = \frac{\ker D \cap \mathrm{im}\, D^{i-1}}{\ker D \cap \mathrm{im}\, D^i}, \qquad 1 \leq i < p.$$

If $\ell_i \in \mathrm{Ver}_p(k)$ is the simple object represented by a Jordan block of length i, then $F(Y) = \bigoplus_{i=1}^{p-1} F_i(Y) \boxtimes \ell_i$. In the semilinear convention, the $F_i(Y)$ are objects of $\mathcal{C}$ itself. F is not assumed exact.

Lemma 6.1 (mod-p identity in K_0)**.** *In $K_0(\mathcal{C})/pK_0(\mathcal{C})$,*

$$[Y]^p = \sum_{i=1}^{p-1} i[F_i(Y)].$$

Consequently, for every $r \geq 1$,

$$[Y]^{p^r} = \sum_{i_1,\ldots,i_r=1}^{p-1} (i_1 \cdots i_r)[F_{i_r} \cdots F_{i_1}(Y)] \pmod{p}. \tag{6.1}$$

Proof. Put $Z = Y^{\otimes p}$ and $K_j = \ker D \cap \mathrm{im}\, D^j$ for $0 \leq j \leq p$. The short exact sequences $0 \to K_j \to \mathrm{im}\, D^j \to \mathrm{im}\, D^{j+1} \to 0$ give $[Z] = \sum_{j=0}^{p-1}[K_j]$. Summing the successive quotients in the filtration $K_0 \supseteq \cdots \supseteq K_p = 0$ gives the integral identity $[Z] = \sum_i i[F_i(Y)] + p[K_{p-1}]$. Mod p this is the first claim. Reraising to successive p-th powers in the commutative ring and applying to concrete objects gives the iterated form. □

6.2 Equality in dimension forces exactness

An exact strong monoidal σ-semilinear functor still preserves Frobenius–Perron dimension: it sends nonzero objects to nonzero objects (from $\mathbf{1} \hookrightarrow X \otimes X^*$), so composing with FP dimension gives a positive normalised character on $K_0(\mathcal{C})$; the FP characterization forces it to be the FP dimension of the source. In particular

$$d(\Phi(X)) = d(X).$$

Lemma 6.2 (first-component inequality)**.** *Let $S \in \mathcal{C}$ be simple and $A \boxtimes B$ a composition factor of $\Phi(S)$. Then $d(A) \leq d(S)$. If equality holds then $\Phi(S) \simeq A \boxtimes B$, B is invertible, and $S \in \mathcal{C}_{\mathrm{ex}}$ (the subcategory on which the ordinary Frobenius functor is exact).*

Proof. Inequality. $d(S) = d(\Phi(S)) = \sum d(A_k)d(B_k) \geq d(A_1)$ for any factor, using $d \geq 1$ for nonzero objects.

Equality. If $d(A) = d(S)$ then the FP decomposition sum is a single term with $d(B) = 1$, so B is invertible and $\Phi(S)$ is simple of the form $A \boxtimes B$.

$S \in \mathcal{C}_{\mathrm{ex}}$. Let Q_A be the projective cover of A. Over k algebraically closed, $\dim_k \mathrm{Hom}(Q_A, X) = [X : A]$. Applying (4.3) to S and $X = S^{\otimes q}$:

$$\Theta(\mathrm{Hom}_{\mathcal{C}}(Q_A, S^{\otimes q})) \cong B.$$

Θ preserves categorical dimension, so its argument has categorical dimension equal to that of B, which is nonzero (its product with the categorical dimension of B^* is one). Hence $[S^{\otimes q} : A] \not\equiv 0 \pmod{p}$.

By (6.1) there are indices $i_1, \ldots, i_r \in \{1, \ldots, p-1\}$ with A a composition factor of Y_r, where $Y_0 = S$, $Y_t = F_{i_t}(Y_{t-1})$. For each t, using $d(F(Y_{t-1})) \geq d(Y_t)$ and $d(\ell_{i_t}) \geq 1$:

$$d(Y_{t-1}) \geq d(F(Y_{t-1})) \geq d(Y_t).$$

But $d(Y_r) \geq d(A) = d(S) = d(Y_0)$, so all inequalities are equalities; in particular $d(F(S)) = d(S)$, so $S \in \mathcal{C}_{\mathrm{ex}}$. □

6.3 The finite graph

Let N be the number of simple isomorphism classes in $\mathcal{C}$. Form a directed graph on those classes by arrow $S \to A$ whenever $A \boxtimes B$ appears in some composition factor of $\Phi(S)$.

Every vertex has at least one outgoing arrow. By lemma 6.2, dimension is nonincreasing along arrows. On a directed cycle, dimension is constant, all vertices belong to $\mathcal{C}_{\mathrm{ex}}$, and each vertex has exactly one outgoing arrow. Hence a path entering a cycle is trapped inside it. A path of length N visits $N + 1$ vertices and so hits a cycle. Its endpoint is in $\mathcal{C}_{\mathrm{ex}}$.

6.4 Iteration and Serre closure

Let $\sigma(a) = a^q$. For a k-linear category $\mathcal{A}$, let $\mathcal{A}^{[\sigma]}$ have the same objects and additive morphism groups but scalar action twisted by σ^{-1}. Then σ-semilinear functors out of $\mathcal{A} = k$-linear functors out of $\mathcal{A}^{[\sigma]}$, and $(\mathcal{A} \boxtimes \mathcal{B})^{[\sigma]} = \mathcal{A}^{[\sigma]} \boxtimes \mathcal{B}^{[\sigma]}$, so products of σ-semilinear functors are σ-semilinear.

Each $\mathrm{Ver}_{p^s}(k)$ is defined over $\mathbb{F}_p$; scalar transport along $b \mapsto b^{p^a}$ is an exact strong symmetric σ^a- semilinear autoequivalence J_{σ^a} of every $\mathrm{Ver}_{p^s}(k)$.

Define $\Psi_0 = \mathrm{Id}$, $\Psi_1 = \Phi$, and

$$\Psi_{j+1} = (\Phi \boxtimes J_\sigma^{\boxtimes j}) \circ \Psi_j : \mathcal{C} \to \mathcal{C} \boxtimes \mathcal{W}^{\boxtimes(j+1)}. \tag{6.2}$$

By induction each Ψ_j is exact, strong symmetric monoidal, and σ^j-semilinear.

Lemma 6.3 (first components after N iterations)**.** *Let $\Psi_N : \mathcal{C} \to \mathcal{C} \boxtimes \mathcal{W}^{\boxtimes N}$ be as above. Then every composition factor $A \boxtimes B_1 \boxtimes \cdots \boxtimes B_N$ of $\Psi_N(X)$ has $A \in \mathcal{C}_{\mathrm{ex}}$.*

Proof. Exactness of Φ plus the path bound on $\mathcal{C}$ (any first component of one iteration is a first component of one further edge); induction on N, starting with simple objects (where the path argument applies) and finishing with composition series. Autoequivalences on other factors send simples to simples and don't touch the first component. □

Let $\mathcal{D} \subseteq \mathcal{C}$ be the full subcategory whose simple composition factors lie in $\mathcal{C}_{\mathrm{ex}}$. This is the *Serre closure* of $\mathcal{C}_{\mathrm{ex}}$.

Lemma 6.4 (Serre closure inherits tensor structure)**.** *$\mathcal{D}$ is closed under tensor products, duals, extensions, and contains the unit; it is finite and has enough projectives; it carries a rigid symmetric tensor structure inherited from $\mathcal{C}$. Moreover the functor Ψ_N factors through*

$$\Psi_N : \mathcal{C} \to \mathcal{D} \boxtimes \mathcal{W}^{\boxtimes N}.$$

Proof. Closed under tensor products: biexactness gives tensor-product filtrations, and $\mathcal{C}_{\mathrm{ex}}$ is a tensor subcategory closed under subquotients (Etingof–Ostrik 2021, Prop 7.4). Duality preserves $\mathcal{D}$ by exactness.

Finiteness, projectives: concretely, write $\mathcal{C} \simeq H$-mod with H basic finite-dimensional; let e be the idempotent summing primitive idempotents of the non-$\mathcal{C}_{\mathrm{ex}}$ simples; then $\mathcal{D} \simeq (H/HeH)$-mod by removing the e-isotypic parts, and H/HeH is finite-dimensional and finite projective dimension, so $\mathcal{D}$ is finite with enough projectives.

Factoring through $\mathcal{D}$: lemma 6.3. □

7 Finishing the proof: a fiber functor on $\mathcal{D}$

7.1 The fiber functor on $\mathcal{D}$

We need an exact strong symmetric monoidal $E : \mathcal{D} \to \mathrm{Ver}_{p^n}(k)$ for some fixed n.

Case $p > 2$. $\mathcal{C}_{\mathrm{ex}}$ is Frobenius-exact, so by Etingof–Ostrik (2021, Theorem 8.1) it has a fiber functor $\mathcal{C}_{\mathrm{ex}} \to \mathrm{Ver}_p(k)$. If $p > 2$, the Serre closure of $\mathcal{C}_{\mathrm{ex}}$ equals $\mathcal{C}_{\mathrm{ex}}$ (Etingof–Ostrik 2021, Corollary 9.5). So $\mathcal{D} = \mathcal{C}_{\mathrm{ex}}$ and we already have $E : \mathcal{D} \to \mathrm{Ver}_p(k)$.

Case $p = 2$. The fiber functor $\mathcal{C}_{\mathrm{ex}} \to \mathrm{Ver}_2(k) = \mathrm{Vec}_k$ is a k-linear fiber functor. But $\mathcal{C}_{\mathrm{ex}} \subseteq \mathcal{D}$ contains every simple of $\mathcal{D}$ by definition, and $\mathcal{C}_{\mathrm{ex}}$ is Frobenius-exact. Etingof–Gelaki (2021, Theorem 2.21(2)) then extends it to $E : \mathcal{D} \to \mathrm{Ver}_4^+(k) \subseteq \mathrm{Ver}_4(k)$. (This is also Etingof–Ostrik 2021, Corollary 9.8.)

So in all cases:

$$E : \mathcal{D} \longrightarrow \mathrm{Ver}_{p^n}(k), \qquad n = \begin{cases} 1 & p > 2, \\ 2 & p = 2. \end{cases}$$

7.2 Combining factors

Set $n = \max(2, r)$ and $\mathcal{Z} = \mathrm{Ver}_{p^n}(k)$. The tower inclusions $\mathrm{Ver}_{p^r}(k) \subseteq \mathrm{Ver}_{p^n}(k)$ (Benson–Etingof–Ostrik 2023, Theorem 4.34) give a k-linear exact strong symmetric monoidal $I : \mathcal{W} \to \mathcal{Z}$.

The functors $E : \mathcal{D} \to \mathcal{Z}$ and $I : \mathcal{W} \to \mathcal{Z}$ combine, by lemma 2.2, into

$$H : \mathcal{D} \boxtimes \mathcal{W}^{\boxtimes N} \to \mathcal{Z}, \qquad H(D \boxtimes W_1 \boxtimes \cdots \boxtimes W_N) = E(D) \otimes I(W_1) \otimes \cdots \otimes I(W_N).$$

H is exact and strong symmetric monoidal.

7.3 Killing the scalar twist

The composite $H\Psi_N : \mathcal{C} \to \mathcal{Z}$ is σ^N-semilinear (each iteration multiplies semilinearity by σ). Let $J_{\sigma^{-N}}$ be scalar transport on $\mathcal{Z}$ along $b \mapsto b^{p^{-N}}$. Then

$$\mathcal{F} := J_{\sigma^{-N}} \circ H \circ \Psi_N : \mathcal{C} \longrightarrow \mathcal{Z}$$

is k-linear, exact, strong symmetric monoidal.

7.4 Faithfulness

Lemma 7.1 (exact strong monoidal with simple unit is faithful)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be an exact strong symmetric monoidal functor between finite symmetric tensor categories. Then F is faithful.*

Proof. Suppose $X \neq 0$. The coevaluation $\mathbf{1} \to X \otimes X^*$ is nonzero (triangle identity) and by simplicity of $\mathbf{1}$ it is injective; applying exact F shows $F(X) \neq 0$. For a nonzero $f : X \to Y$: exactness gives $\operatorname{im} F(f) \cong F(\operatorname{im} f)$, and $\operatorname{im} f \neq 0$, so $F(f) \neq 0$. □

Applying lemma 7.1 to $\mathcal{F}$ shows $\mathcal{F}$ is faithful. Hence $\mathcal{F} : \mathcal{C} \to \mathrm{Ver}_{p^n}(k)$ is the required fiber functor, and theorem 1.4 follows.

8 Consequences

We now record the three structural corollaries that were the original motivation for proving theorem 1.4.

8.1 Incompressible categories

Recall that $\mathcal{C}$ is *incompressible* if every k-linear exact strong symmetric monoidal functor from $\mathcal{C}$ to a symmetric tensor category over k is fully faithful.

Corollary 8.1 (classification of finite incompressible)**.** *Let k be algebraically closed of characteristic $p > 0$. A finite symmetric tensor category $\mathcal{C}$ over k is incompressible if and only if it is k-linearly equivalent as a symmetric tensor category to one of the following, according to p:*

$$\begin{array}{l|l} p = 2 & \mathrm{Ver}_{2^m}(k)\ (m \geq 1), \quad \mathrm{Ver}^+_{2^m}(k)\ (m \geq 2), \\ p = 3 & \mathrm{Ver}_{3^m}(k),\ \mathrm{Ver}^+_{3^m}(k) \quad (m \geq 1), \\ p \geq 5 & \mathrm{Vec}_k,\ \mathrm{sVec}_k, \quad \mathrm{Ver}_{p^m}(k),\ \mathrm{Ver}^+_{p^m}(k) \quad (m \geq 1). \end{array}$$

Proof. Direction $\Leftarrow$: Vec_k has trivial $\mathrm{End}(\mathbf{1}) = k$, so any strong symmetric tensor functor out of it is fully faithful (fill in the linear maps on simples); the supervector-space case is similar (an odd simple has a nontrivial sign in its double dual, which cannot match a vector space). For the Verlinde full and plus categories, see Benson–Etingof– Ostrik (2023), Theorems 4.71 and 4.72(iii).

Direction $\Rightarrow$: take a fiber functor $F : \mathcal{C} \to \mathrm{Ver}_{p^n}(k)$ from theorem 1.4, let $\mathcal{I}$ be the image's full subcategory of subquotients. Then F corestricts to a surjective-in-the-tensor-category-sense functor onto $\mathcal{I}$ (Etingof–Gelaki–Nikshych–Ostrik 2015, Definition 1.8.18, Prop 1.8.19, Section 6.3). Incompressibility of $\mathcal{C}$ forces this corestriction to be fully faithful. The two FP-dimension inequalities of finiteness (inclusion vs surjection) then force equality, hence an equivalence. Benson–Etingof–Ostrik 2023, Corollary 4.61 then restricts $\mathcal{I}$ to a full or plus sub-Verlinde at height $\leq n$. Collecting cases gives the stated list. □

Every finite tensor category has moderate growth: multiplication by the class of a fixed object on the finite-rank Grothendieck group is a finite nonnegative matrix, which bounds the lengths of its tensor powers exponentially. So corollary 8.1 is precisely the finite case of Conjecture B of [CEO2024].

8.2 Uniqueness in characteristic 2

Corollary 8.2 (uniqueness in characteristic two)**.** *Let k be algebraically closed of characteristic two and $\mathcal{C}$ a finite symmetric tensor category. Then $\mathcal{C} \to \mathrm{Ver}_{2^\infty}(k) := \bigcup_n \mathrm{Ver}_{2^n}(k)$ has exactly one fiber functor up to monoidal natural isomorphism; its class contains a representative into a finite stage $\mathrm{Ver}_{2^n}(k)$; and for each fixed n there is at most one fiber functor $\mathcal{C} \to \mathrm{Ver}_{2^n}(k)$ up to monoidal natural isomorphism.*

Proof. Existence: theorem 1.4 into some finite stage, then tower inclusion. Uniqueness (union): Coulembier–Etingof–Ostrik (2024), Theorem 9.4.1: any k-linear exact symmetric tensor functor from an arbitrary tensor category in characteristic 2 into $\mathrm{Ver}_{2^\infty}(k)$ is unique up to monoidal natural isomorphism. Fixed stage: lift stage uniqueness through the fully faithful tower inclusion; naturality and tensor compatibility hold by faithfulness. □

8.3 Tannaka reconstruction over a Verlinde level

Let $\mathcal{V}$ be a finite symmetric tensor category. The fundamental group $\pi(\mathcal{V}) := \underline{\mathrm{Aut}}^{\otimes}(\mathrm{Id}_{\mathcal{V}})$ acts canonically on $\mathrm{Ind}(\mathcal{V})$. A *$\mathcal{V}$-group* is an internal affine group scheme G with a homomorphism $\phi : \pi(\mathcal{V}) \to G$ whose induced coaction matches the canonical one.

Corollary 8.3 (reconstruction over a Verlinde level)**.** *Let k be algebraically closed of characteristic $p > 0$, $\mathcal{C}$ finite symmetric tensor, and $F : \mathcal{C} \to \mathcal{V} = \mathrm{Ver}_{p^n}(k)$ a fiber functor (from theorem 1.4). Then there is a $\mathcal{V}$-group (G, ϕ) with $\mathcal{C} \simeq \mathrm{Rep}(G, \phi)$ over $\mathcal{V}$, equivalent to F under the forgetful functor. One may take $G = \underline{\mathrm{Aut}}^{\otimes}(F)$ and $\phi : \pi(\mathcal{V}) \to \underline{\mathrm{Aut}}^{\otimes}(F)$ the canonical homomorphism.*

Proof. Lemma 2.1 makes $\mathcal{C}$, together with $\mathcal{V} = \mathrm{Ver}_{p^n}(k)$, a "pretannakian category" in the sense of Coulembier–Etingof–Ostrik (2024). The chosen F is a tensor functor, and their Theorem 4.2.1 produces the stated $\mathcal{V}$-group; conversely, forgetting G-structure gives the inverse functor. Composition is the equivalence. □

Closing remarks

All classification and existence assertions above are confined to finite categories. Uniqueness is asserted only in characteristic two, both in the tower union and at each fixed finite stage where a fiber functor exists. Theorem 1.4 supplies some finite height n depending on $\mathcal{C}$; it makes *no* claim of existence at a prescribed height. In the reconstruction, G is affine but neither its finiteness nor its finite-type property is used or claimed.

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